

EXCEL DATA MODELLING PROJECT

Project overview:

This project demonstrates the ability to transform raw, fragmented data into a structured relational model. Using excel data modelling project.xlsx, I established complex connections between multiple tables to facilitate deep-dive analysis and automated reporting.

Key Features

- **Relational Data Modelling:** Created a "Star Schema" by linking dimension tables to fact tables using Power Pivot.
- **DAX Measures:** Authored custom Data Analysis Expressions (DAX) to calculate advanced metrics and KPIs.
- **Data Integrity:** Implemented data cleaning and normalization techniques to ensure accuracy across 8 integrated sheets.
- **Scalability:** Designed the model to handle large datasets efficiently without compromising performance.

Tech Stack

- Microsoft Excel (Power Pivot, Power Query)
- DAX (Data Analysis Expressions)

SCREENSHOTS OF THE ABOVE PROJECT:

	Customer	Gender	Name	City	State Code	State	Zip Code	Country	Birthday	Age	Age Group
Customer Data	3001	Female	Lilly Harding	Wandearah East	Sa	South Australia	5523	Australia	07-03-1939	87	50+
	325	Female	Madison Hull	Mount Budd	Wa	Western Australia	6522	Australia	27-09-1979	46	35-49
	354	Female	Claire Fernes	Winjallock	Vic	Victoria	3380	Australia	26-05-1947	79	50+
Exchange Rate Data	786	Male	Jai Poltpalingada	Middle River	Sa	South Australia	5223	Australia	17-09-1957	69	50+
	1042	Male	Aidan Rankhurst	Tawonga South	Vic	Victoria	3686	Australia	19-11-1965	60	50+
	1086	Male	Hayden Clegg	Templers	Sa	South Australia	5371	Australia	20-01-1954	72	50+
Products Data	1133	Male	Nicholas Caffyn	Jubilee Pocket	Qld	Queensland	4802	Australia	22-11-1969	56	50+
	1256	Male	Lincoln Jenks	Kulligum	Qld	Queensland	4660	Australia	03-12-1950	75	50+
	1314	Male	Isaac Israel	Edith River	Nt	Northern Territory	852	Australia	21-12-1965	60	50+
Sales Data	1568	Male	Luke Virtue	Korta	Vic	Victoria	3565	Australia	25-07-1975	51	50+
	1585	Female	Anna Hallstrom	Green Lake	Vic	Victoria	3401	Australia	08-12-1990	35	35-49
	1626	Female	Mary Leach	Needs Corner	Vic	Victoria	3496	Australia	17-10-1985	40	35-49
Stores Data	1642	Female	Ruby Ambrose	Tarlo	Nsw	New South Wales	2580	Australia	03-03-1965	61	50+
	1677	Male	Dean Innes	Ellesmere	Qld	Queensland	4610	Australia	22-11-1964	61	50+
	1817	Female	Laura Le Rennetel	Beauchamp	Vic	Victoria	3579	Australia	07-10-1986	39	35-49
Objectives	1963	Female	Chelsea Watkins	Nambour Dc	Qld	Queensland	4590	Australia	22-09-1984	41	35-49
	1945	Male	Angus Beaurepaine	Greenfields	Wa	Western Australia	6210	Australia	20-09-1974	51	50+
	2238	Female	Isabelle Cochran	Brucknell	Vic	Victoria	3268	Australia	31-10-1969	56	50+
Analysis	2248	Male	Rory Spargo	Jung	Vic	Victoria	3401	Australia	15-12-1987	38	35-49
	2435	Female	Lillian Hall	Blakebrook	Nsw	New South Wales	2480	Australia	04-01-1968	58	50+
	2469	Female	Phoebe Kneeshaw	Bidwill	Qld	Queensland	4630	Australia	20-05-1948	78	50+
Revenue & Profit DB	2512	Female	Matilda Hake	Bunkers Hill	Vic	Victoria	3352	Australia	13-11-1992	33	20-34
	2604	Female	Claudia Davison	Kubin Village	Qld	Queensland	4875	Australia	04-08-1995	31	20-34
	2751	Male	Nate Loves	Callide	Qld	Queensland	4715	Australia	20-08-1979	47	35-49
Product	2792	Male	Zachary Perkin	Bamgarie	Vic	Victoria	3333	Australia	25-08-1976	50	50+
	3036	Male	Nicholas Shawnd	Gordonbrook	Qld	Queensland	4610	Australia	23-02-1959	67	50+
	3002	Male	Joshua Anonson	Smithton	Tas	Tasmania	7530	Australia	11-08-2000	26	20-34
	3006	Female	Alicia Stuckey	Hartley	Nsw	New South Wales	2790	Australia	13-01-1943	83	50+
	3203	Male	Aidan Kane	Argenton	Nsw	New South Wales	2284	Australia	18-08-1998	28	20-34
	3270	Male	Hunter Templeton	Central Tilba	Nsw	New South Wales	2546	Australia	09-02-1948	78	50+
	3327	Female	Aleimah Le Messurier	Kongroovootong	Vic	Victoria	3315	Australia	28-04-1948	78	50+
	3410	Male	Will Armistage	Bollon	Qld	Queensland	4488	Australia	05-12-1950	75	50+
	3575	Male	Noah McFarland	Hoppers Crossing	Vic	Victoria	3029	Australia	03-04-1942	84	50+
	3964	Female	Rose Code	Dixie	Vic	Victoria	3265	Australia	16-02-1944	82	50+
	4004	Female	Piper Writson	Wyalla Plaza	Qld	Queensland	4350	Australia	24-06-1995	31	20-34

	Date	Currency	Exchange
Customer Data	01-01-2015	USD	1.00
	01-01-2015	CAD	1.16
	01-01-2015	AUD	1.22
Exchange Rate Data	01-01-2015	EUR	0.82
	01-01-2015	GBP	0.64
	01-02-2015	USD	1.00
Products Data	01-02-2015	CAD	1.17
	01-02-2015	AUD	1.21
	01-03-2015	EUR	0.83
Sales Data	01-02-2015	GBP	0.65
	01-03-2015	USD	1.00
	01-03-2015	CAD	1.17
Stores Data	01-03-2015	AUD	1.21
	01-03-2015	EUR	0.83
	01-03-2015	GBP	0.65
Objectives	01-04-2015	USD	1.00
	01-04-2015	CAD	1.17
	01-04-2015	AUD	1.21
Analysis	01-04-2015	EUR	0.83
	01-04-2015	GBP	0.65
	01-05-2015	USD	1.00
Revenue & Profit DB	01-05-2015	CAD	1.18
	01-05-2015	AUD	1.24
	01-05-2015	EUR	0.84
Product	01-05-2015	GBP	0.66
	01-06-2015	USD	1.00
	01-06-2015	CAD	1.18
	01-06-2015	AUD	1.21
	01-06-2015	EUR	0.84
	01-06-2015	GBP	0.66
	01-07-2015	USD	1.00
	01-07-2015	CAD	1.18
	01-07-2015	AUD	1.24
	01-07-2015	EUR	0.83
	01-07-2015	GBP	0.66

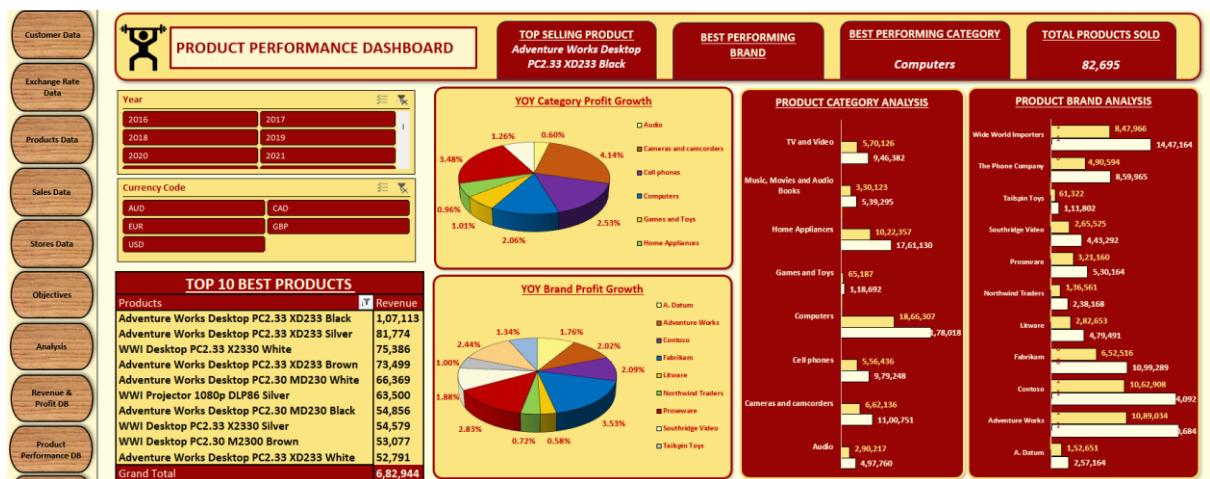
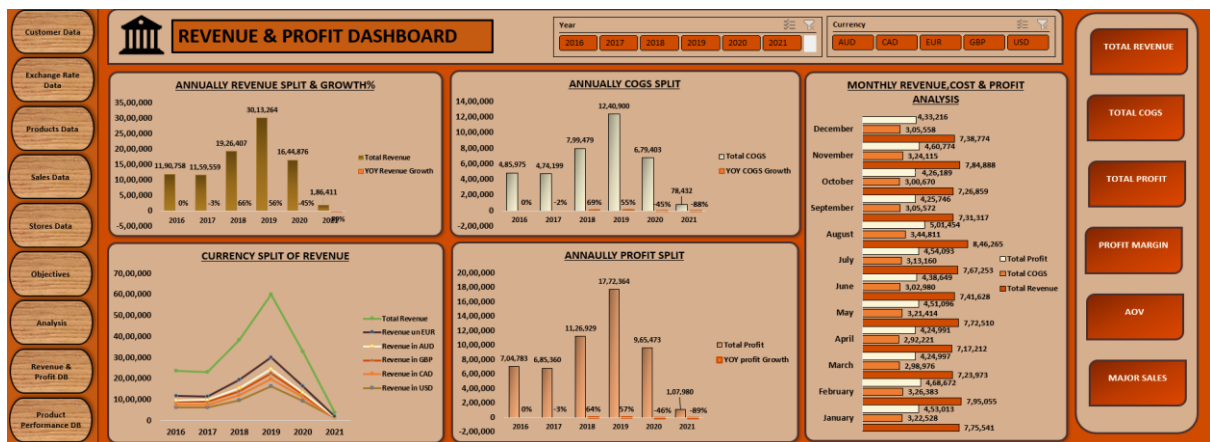
	ProductKey	Product Name	Brand	Color	Unit Cost USD	Unit Price USD	SubcategoryKey	Subcategory	CategoryKey	Category
Customer Data	1	Contoso 32GB MP3 Player ES1 Silver	Contoso	Silver	6.82	12.99	101	MP4&MP3	1	Audio
	2	Contoso 32GB MP3 Player ES1 Blue	Contoso	Blue	6.82	12.99	101	MP4&MP3	1	Audio
	3	Contoso 16 MP3 Player ES1 White	Contoso	White	14.52	24.99	101	MP4&MP3	1	Audio
	4	Contoso 2G MP3 Player E200 Silver	Contoso	Silver	11.00	21.57	101	MP4&MP3	1	Audio
Exchange Rate Data	5	Contoso 2G MP3 Player E200 Red	Contoso	Red	11.00	21.57	101	MP4&MP3	1	Audio
	6	Contoso 2G MP3 Player E200 Black	Contoso	Black	11.00	21.57	101	MP4&MP3	1	Audio
	7	Contoso 2G MP3 Player E200 Blue	Contoso	Blue	11.00	21.57	101	MP4&MP3	1	Audio
	8	Contoso 4G MP3 Player E400 Silver	Contoso	Silver	30.58	59.99	101	MP4&MP3	1	Audio
Products Data	9	Contoso 4G MP3 Player E400 Black	Contoso	Black	30.58	59.99	101	MP4&MP3	1	Audio
	10	Contoso 4G MP3 Player E400 Green	Contoso	Green	30.58	59.99	101	MP4&MP3	1	Audio
	11	Contoso 4G MP3 Player E400 Orange	Contoso	Orange	30.58	59.99	101	MP4&MP3	1	Audio
	12	Contoso 4GB Flash MP3 Player E401 Blue	Contoso	Blue	35.72	77.68	101	MP4&MP3	1	Audio
Sales Data	13	Contoso 4GB Flash MP3 Player E401 Black	Contoso	Black	35.72	77.68	101	MP4&MP3	1	Audio
	14	Contoso 4GB Flash MP3 Player E401 Silver	Contoso	Silver	35.72	77.68	101	MP4&MP3	1	Audio
	15	Contoso 4GB Flash MP3 Player E401 White	Contoso	White	35.72	77.68	101	MP4&MP3	1	Audio
	16	Contoso 8GB Super-Slim MP3/Video Player M800 White	Contoso	White	109.56	199.99	101	MP4&MP3	1	Audio
Stores Data	17	Contoso 8GB Super-Slim MP3/Video Player M800 Red	Contoso	Red	50.56	109.95	101	MP4&MP3	1	Audio
	18	Contoso 8GB Super-Slim MP3/Video Player M800 Green	Contoso	Green	50.56	109.95	101	MP4&MP3	1	Audio
	19	Contoso 8GB Super-Slim MP3/Video Player M800 Pink	Contoso	Pink	50.56	109.95	101	MP4&MP3	1	Audio
	20	Contoso 8GB MP3 Player new model M200 Black	Contoso	Black	61.82	134.00	101	MP4&MP3	1	Audio
Objectives	21	Contoso 8GB MP3 Player new model M200 Blue	Contoso	Blue	61.82	134.00	101	MP4&MP3	1	Audio
	22	Contoso 8GB MP3 Player new model M200 Yellow	Contoso	Yellow	61.82	134.00	101	MP4&MP3	1	Audio
	23	Contoso 8GB MP3 Player new model M200 White	Contoso	White	61.82	134.00	101	MP4&MP3	1	Audio
	24	Contoso 16GB MP3 Player M1600 Blue	Contoso	Blue	91.93	199.99	101	MP4&MP3	1	Audio
Analysis	25	Contoso 16GB MP3 Player M1600 Black	Contoso	Black	91.93	199.99	101	MP4&MP3	1	Audio
	26	Contoso 16GB MP3 Player M1600 Green	Contoso	Green	91.93	199.99	101	MP4&MP3	1	Audio
	27	Contoso 16GB MP3 Player M1600 White	Contoso	White	91.93	199.99	101	MP4&MP3	1	Audio
	28	Contoso 16GB MP3 Player M1600 Red	Contoso	Red	91.93	199.99	101	MP4&MP3	1	Audio
Revenue & Profit DB	29	Contoso 32GB Video MP3 Player M3200 White	Contoso	White	84.49	255.00	101	MP4&MP3	1	Audio
	30	Contoso 32GB Video MP3 Player M3200 Red	Contoso	Red	84.49	255.00	101	MP4&MP3	1	Audio
	31	Contoso 32GB Video MP3 Player M3200 Orange	Contoso	Orange	84.49	255.00	101	MP4&MP3	1	Audio
	32	Contoso 32GB Video MP3 Player M3200 Pink	Contoso	Pink	84.49	255.00	101	MP4&MP3	1	Audio
Product	33	Contoso 32GB Video MP3 Player M3200 Black	Contoso	Black	84.49	255.00	101	MP4&MP3	1	Audio
	34	Contoso 4GB Portable MP3 Player M400 Black	Contoso	Black	95.95	95.95	101	MP4&MP3	1	Audio
	35	Contoso 4GB Portable MP3 Player M400 White	Contoso	White	48.92	95.95	101	MP4&MP3	1	Audio

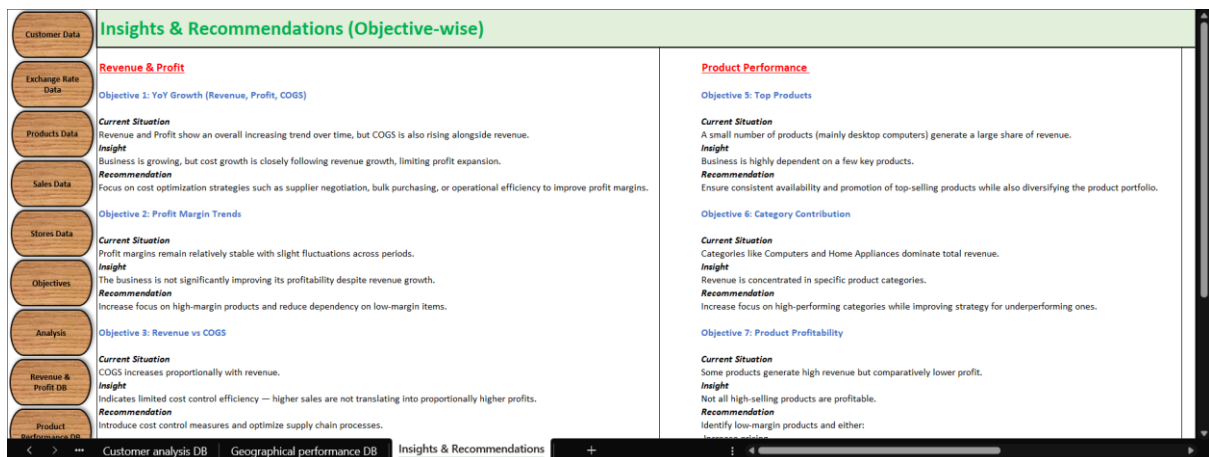
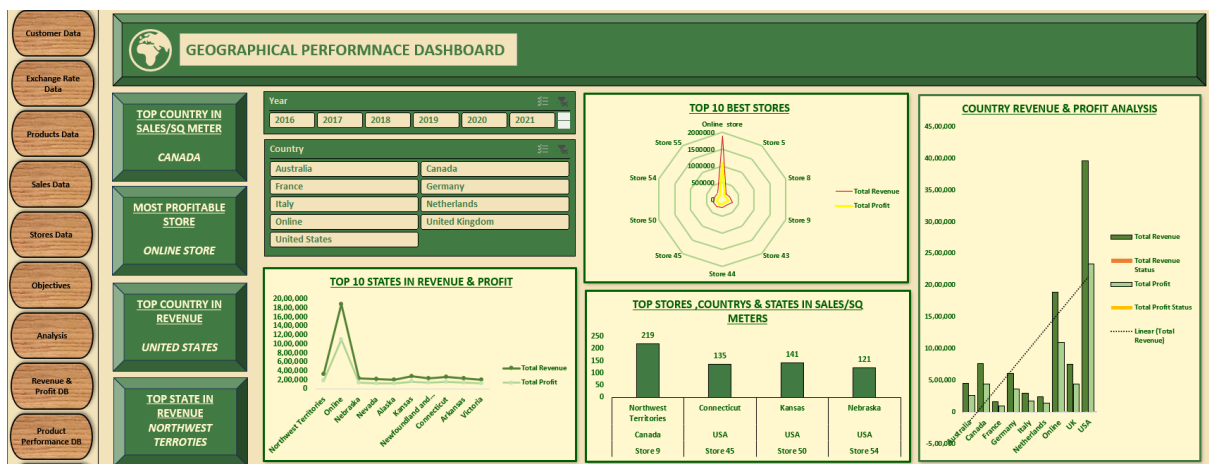
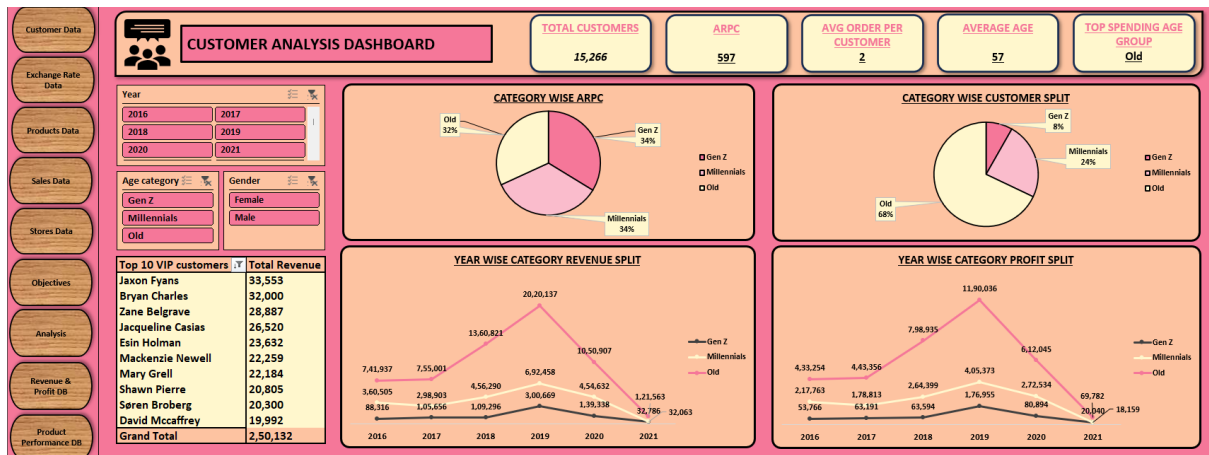
	Order Number	Line Item	Order Date	Delivery Date	Delivery Status	Customer Key	Store Key	Product Key	Quantity	Currency Code
Customer Data	366000	1	01-01-2016		In Transit	265598	10	1304	1	CAD
	366001	1	01-01-2016	13-01-2016	Delivered	1269051	0	1048	2	CAD
	366002	1	01-01-2016	01-12-2016	Delivered	266019	0	1106	2	CAD
Exchange Rate Data	366004	1	01-01-2016		In Transit	1107461	38	163	6	GBP
	366005	1	01-01-2016		In Transit	844009	33	421	4	EUR
	366007	1	01-01-2016		In Transit	2013771	43	1617	1	USD
Products Data	366008	1	01-01-2016		In Transit	759705	29	1154	9	EUR
	366009	1	01-01-2016		In Transit	254540	9	1648	1	CAD
	366010	1	01-01-2016	01-08-2016	Delivered	170077	0	618	2	CAD
Sales Data	366011	1	01-01-2016		In Transit	1984955	66	128	7	USD
	366012	1	01-01-2016		In Transit	3977527	65	1807	1	USD
	366013	1	01-01-2016		In Transit	1537009	41	1654	5	GBP
Stores Data	366014	1	01-01-2016	01-06-2016	Delivered	778549	0	123	1	USD
	366016	1	01-01-2016		In Transit	1982762	63	1253	7	USD
	367001	1	01-02-2016		In Transit	1438058	56	1176	1	USD
Objectives	367003	1	01-02-2016		In Transit	901164	40	153	2	GBP
	367004	1	01-02-2016		In Transit	1853699	56	136	1	USD
	367005	1	01-02-2016	01-10-2016	Delivered	756280	0	1691	1	EUR
Analysis	367006	1	01-02-2016		In Transit	638285	18	2445	1	USD
	367007	1	01-02-2016		In Transit	1730985	57	65	7	USD
	367008	1	01-02-2016		In Transit	761246	28	1951	3	EUR
Revenue & Profit DB	367009	1	01-02-2016		In Transit	1612558	57	934	1	USD
	367010	1	01-02-2016		In Transit	1322959	66	917	1	USD
	367011	1	01-02-2016		In Transit	507981	22	19	1	USD
Product	367012	1	01-02-2016		In Transit	2099996	9	1653	2	USD
	367013	1	01-02-2016		In Transit	815458	31	1733	2	EUR
	367014	1	01-02-2016		In Transit	1839627	56	1164	3	USD
Revenue & Profit DB	367015	1	01-02-2016		In Transit	911025	42	2296	4	GBP
	367016	1	01-02-2016		In Transit	6013409	63	1487	1	USD
	367017	1	01-02-2016		In Transit	1612400	47	1203	4	GBP
Product	367018	1	01-02-2016		In Transit	1301565	29	1285	2	EUR
	367019	1	01-02-2016		In Transit	793573	29	1947	2	EUR
	367020	1	01-02-2016		In Transit	799366	29	448	6	EUR
Product	367021	1	01-02-2016		In Transit	2419117	10	1445	4	CAD
	367022	1	01-02-2016		In Transit	942807	42	1596	1	GBP

	Storekey	Country	State	Square Meters	Open Date
Customer Data	1	Australia	Australian Capital Territory	595	01-01-2008
	2	Australia	Northern Territory	665	01-12-2008
	3	Australia	South Australia	2000	01-07-2012
	4	Australia	Tasmania	2000	01-01-2010
Exchange Rate Data	5	Australia	Victoria	2000	12-09-2015
	6	Australia	Western Australia	2000	01-01-2010
	7	Canada	New Brunswick	1105	09-07-2007
Products Data	8	Canada	Newfoundland and Labrador	2103	07-03-2014
	9	Canada	Northwest Territories	1500	03-04-2005
	10	Canada	Nunavut	1210	04-04-2015
	11	Canada	Nukun	1210	06-09-2009
Sales Data	12	France	Basse Normandie	350	06-06-2012
	13	France	Corse	245	06-07-2013
	14	France	Franchise-Com&G	350	15-12-2009
	15	France	La Réunion	400	01-03-2015
Stores Data	16	France	Limousin	385	06-03-2010
	17	France	Martinique	350	07-08-2007
	18	France	Mayotte	310	08-08-2012
Objectives	19	Germany	Berlin	1295	04-04-2013
	20	Germany	Brandenburg	1715	15-12-2012
	21	Germany	Freie Hansestadt Bremen	560	06-03-2018
	22	Germany	Freistaat Thüringen	2000	03-06-2008
Analysis	23	Germany	Hamburg	1385	01-01-2010
	24	Germany	Hessen	1855	15-12-2012
	25	Germany	Mecklenburg-Vorpommern	1610	01-01-2010
	26	Germany	Saarland	350	03-05-2015
Revenue & Profit DB	27	Germany	Sachsen-Anhalt	2000	08-08-2008
	28	Italy	Calanissetta	1200	15-12-2012
	29	Italy	Enna	1000	01-01-2008
Product	30	Italy	Pesaro	2100	01-12-2009
	31	Netherlands	Drenthe	1085	01-07-2012
	32	Netherlands	Flevoland	910	01-01-2010
	33	Netherlands	Friesland	1540	12-09-2015
	34	Netherlands	Groningen	1165	01-01-2010
	35	Netherlands	Zeeland	1225	05-07-2007

OBJECTIVES		
Customer Data	REVENUE AND PROFIT BASED OBJECTIVES	
	Objective 1 To analyze Year-over-Year (YoY) growth in Revenue, Profit, and COGS (%) Purpose: To understand business growth trends and cost behavior over time.	
	Objective 2 To evaluate Profit Margin trends over time Purpose: To identify whether profitability is improving or declining.	
	Objective 3 To compare Revenue vs COGS to assess cost efficiency Purpose: To determine how effectively the company is controlling costs.	
	Objective 4 To identify seasonal trends in monthly sales performance Purpose: To detect peak and low demand periods.	
	Product Performance Based Objective Objective 5 To identify top-performing products based on Revenue and Profit Purpose: To determine which products drive the majority of business value.	
Exchange Rate Data	Objective 6 To analyze category-wise contribution to total revenue Purpose: To understand which product categories dominate sales.	
	Objective 7 To evaluate product-level profitability Purpose: To identify high-revenue but low-profit products.	
	Customer Based Objective Objective 8 To analyze customer distribution by age group and its impact on revenue Purpose: To identify the most valuable customer segment.	
	Objective 9 To compare purchasing behavior across gender Purpose: To check if gender significantly influences sales.	
	Objective 10 To evaluate customer contribution to overall revenue Purpose: To understand whether revenue is concentrated among specific customer groups.	
Products Data	Customers	Exchange rate
Sales Data	Products	Sales
Stores Data	Stores	Objectives
Objectives	Analysis	Revenue & profit DB

REVENUE AND PROFIT ANALYSIS										PRODUCT PERFORMANCE																																																												
Customer Data	KPI Cards										KPI Cards																																																											
	Total Revenue		Total COGS		Total Profit		Profit Margin		AOV		Major sales chann.		Total Revenue		Total Revenue		Total Revenue																																																					
Exchange Rate Data	91,21,276										37,58,387										53,62,889		58.80%		346		Online		18,86,766		1,07,113																																							
	Revenue from every currency										Revenue from every currency										Revenue from every currency										Revenue from every currency																																							
Products Data	Year										Revenue in USD										Revenue in CAD										Revenue in GBP										Revenue in AUD										Revenue in EUR										Total Revenue									
	2016										8,29,807										8,36,117										8,56,211										85,373										8,38,720										11,90,758									
Sales Data	2017										8,36,987										8,32,098										8,53,615										79,810										8,78,049										11,59,559									
	2018										8,77,145										8,22,906										2,04,625										8,47,943										9,79,788										19,26,407									
Stores Data	2019										8,66,879										8,15,575										8,67,456										8,24,907										8,56,654										30,13,264									
	2020										8,85,196										8,35,106										86,429										8,37,330										8,64,478										31,78,018									
Objectives	2021										8,85,890										8,43,337										8,15,857										8,125										8,202										1,86,411									
	Grand Total										49,37,426										10,00,980										9,24,459										5,86,288										16,62,124										91,21,276									
Analysis	Monthly revenue Cost & profit analysis										Total Revenue										Total COGS										Total Profit										Field										Representing colour																			
	January										7,75,541										2,72,528										4,53,013										Revenue										COGS																			
Revenue & Profit DB	February										7,75,081										2,72,528										4,53,013										COGS										Profit																			
	March										7,75,973										2,72,576										4,53,997										Profit										Profit																			
Product Performance DB	April										7,77,212										2,72,251										4,54,991										COGS										Profit																			
	May										7,75,510										2,72,454										4,53,096										COGS										Profit																			
Customers	June										7,62,008										2,72,860										4,53,608										COGS										Profit																			
	July										7,67,253										2,71,130										4,54,093										COGS										Profit																			
Exchange rate	August										8,46,265										2,64,811										5,01,454										COGS										Profit																			
	September										8,31,317										2,66,572										4,25,768										COGS										Profit																			
Products	October										8,26,359										2,66,816										4,25,768										COGS										Profit																			
	November										7,84,888										2,64,115										4,60,774										COGS										Profit																			
Stores	December										7,98,774										2,60,558										4,93,216										COGS										Profit																			
	Grand Total										91,21,276										37,58,387										53,62,889										COGS										Profit																			
Sales	Yearly total revenue & growth										Total Revenue										YOY Revenue Growth										Field										Representing colour																													
	2016										8,29,807										8,36,117										8,56,211										85,373										8,38,720										11,90,758									
Objectives	2017										8,36,987										8,32,098										8,53,615										79,810										8,78,049										11,59,559									
	2018										8,77,145										8,22,906										2,04,625										8,47,943										9,79,788										19,26,407									
Analysis	2019										8,66,879										8,15,575										8,67,456										8,24,907										8,56,654										30,13,264									
	2020										8,85,196										8,35,106										86,429										8,37,330										8,64,478										31,78,018									
Revenue & Profit DB	2021										8,85,890										8,43,337										8,15,857										8,125										8,202										1,86,411									
	Grand Total										49,37,426										10,00,980										9,24,459										5,86,288										16,62,124										91,21,276									
Product	Top 10 Products										Total Revenue										Total Profit										Total																																							
	WWI Projector 1080p DLP86 Silver										6,500										1,07,113										2,98,217																																							
Customers	Adventure Works Desktop PC2.33 XD233 Black										71,499										18,86,766										1,07,113																																							
	Adventure Works Desktop PC2.33 XD233 Brown										81,774										18,86,766										1,07,113																																							
Exchange rate	WWI Desktop PC2.33 X2330 Silver										54,579										18,86,766										1,07,113																																							
	WWI Desktop PC2.33 X2330 White										73,386										18,86,766										1,07,113																																							
Products	Adventure Works Desktop PC2.33 MD230 Black										54,856										18,86,766										1,07,113																																							
	Adventure Works Desktop PC2.33 MD230 White										66,369										18,86,766										1,07,113																																							
Stores	Adventure Works Desktop PC2.33 XD233 White										53,791										18,86,766										1,07,113																																							
	WWI Desktop PC2.33 X2330 Brown										53,072										18,86,766										1,07,113																																							
Objectives	Customers										Exchange rate										Products										Sales										Stores										Objectives																			
	Analysis										Revenue & profit DB										++										+										+																													





THANK YOU!